A GROWING ARMY

FIVE OF THE 100-ODD PRIVATE START-UPs COMPETING FOR INDIA'S SPACE SECTOR

SKYROOT AEROSPACE

FOUNDED: 2018; FOUNDERS: Pawan Chandana, Bharath Daka

Skyroot provides launch services for small satellites. It successfully tested its first launcher Vikram-S on November 18; as of 2022, it has completed full ground tests of its first rocket, Vikram I. Two heavier rockets—Vikram II and Vikram III—are currently in development.

AGNIKUL COSMOS

FOUNDED: 2017; FOUNDERS: Srinath Ravichandran, Moin S.P.M.

Developing and testing its spacefaring technology since 2018, AgniKul successfully developed a single-piece fully 3D printed upper stage engine in 2019. Its first launcher is expected to be tested in December.

DHRUVA SPACE

FOUNDED: 2012; FOUNDERS: Sanjay Nekkanti, Krishna Teja Penamakuru, Abhay Egoor, Chaitanya Dora Surapureddy

The firm was established to lead the privatisation of the satellite industry in India. It says it is developing an indigenous small satellite platform with a focus on assembly, integration, testing and operation of satellites.

PIXXEL

FOUNDED: 2019; FOUNDERS: Awais Ahmed, Kshitij Khandelwal

The firm describes itself as a space data company, building a stable of hyper-spectral Earth-imaging satellites and the analytical tools needed to mine insights from the data they gather.

BELLATRIX AEROSPACE

FOUNDED: 2015; FOUNDERS: Rohan M. Ganapathy, Yashas Karanam

The firm specialises in in-space propulsion systems and orbital launch vehicles. It works to facilitate the development of key technologies in electrical propulsion, new-generation propellants and launch vehicles.

the space sector that permitted Indian private players to not only build rockets and satellites but launch them as well. It also mandated that ISRO allow these private companies to use its facilities to test and launch their rockets and other space-based activities. To regulate, coordinate and facilitate all private sector launches and satellite-building across the country, the government formed the Indian National Space Promotion and Authorisation Centre (IN-SPACe) the same year. The reforms also gave a boost to the plans for NewSpace India Limited, a public sector unit that was set up in 2019, to enter into joint ventures with Indian industry to produce, assemble and integrate launch vehicles initially and satellites later on. "These path-breaking initiatives were designed to keep the doors wide open for the private industry, particularly start-ups, in the space sector. In fact, we should have done it decades earlier," says MoS Singh. For Skyroot, the development was fortuitous as they found ISRO both cooperative and responsive to their needs.

Equally challenging was overcoming the technology obstacles. For, as Chandana put it, "Rocket science is rocket science." Skyroot did not want to reinvent the wheel. Instead, the rocket, which they named after Vikram Sarabhai, the father of the Indian space programme, used several cutting-edge technologies that they could claim as the first in India. These included using carbon-fibre for its motor casing that helps the rocket be lighter but stronger. They also employed the latest 3-D printing technology to build its thrusters. Despite the delays caused by the Covid pandemic, they surprised industry-watchers by the speed with which they readied the rocket for launch. They put Vikram-S through several major tests to ensure it would perform flawlessly and even had a top ISRO group validate all the systems.

Yet, on the day of the launch, Chandana was worried. He knew that space was unforgiving and even the slightest error would mean the end of four years of their work. As the countdown hit zero, Vikram-S lifted off with a blaze of bright orange fumes and smoke on its tail. Within 20 seconds, it had